

1. Describe two pros and two cons of SLMs compared to LLMs. [2 points]

SLMs are useful when:

- Privacy is critical and edge device deployment is needed
- Need for fast models (lower inference time)

However, SLMs:

- Are often found to do well in narrow tasks and struggle with broader range of tasks (partly due to reduced model size - parameters)
- Soft Label Accessibility: Often, one needs soft labels from LLMs (which is proprietary) to train SLMs

2. List 2-4 different measures you would use for evaluating quality of an LM's generation. Write a sentence for each explaining what it is or what it's used for. [2 points]

BLEU measures the overlap of n-grams between the model-generated text and a reference text, commonly used in machine translation to assess how similar the generated output is to human-written translations. Higher the better.

ROUGE evaluates the quality of generated text by comparing it to reference texts using recall-based measures of overlapping n-grams, often used in summarization tasks to assess how much key content was captured.

Perplexity measures how well a language model predicts the next word in a sequence; lower perplexity indicates the model is more confident and accurate in its predictions, mainly used for evaluating language models during training or on benchmark datasets. Lower the better.

3. In the class we started building a pipeline that uses an SLM for text generation, but there were imperfections. Specifically, we got more than what we asked for as the SLM continued generating text beyond the answer. Figure out at least one way to fix this. Yes, this will be an engineering/coding based solution, but that's alright. In your response, explain what your solution is and present perplexity scores for the original solution and your solution for at least 5 different prompts. [6 points]

Firstly, the model continues generating text because of the structure of the training data. The model will stop generating only when the next token's prob for <EOS> is max. But due to limited training on diverse data, that's not happening. This generally happens when the model has been trained on only specific type of data (FAQs, QnA etc).

A quick fix is to add a code block which splits responses based on special hardcoded keywords.

```
generated_tokens = outputs[0][input_length:]
generated_text = tokenizer.decode(generated_tokens, skip_special_tokens=True)

# Post-process to stop at first line break or special marker
stop_phrases = ['Exercise', 'Question', 'Note:']
for stop_phrase in stop_phrases:
    if stop_phrase in generated_text:
        generated_text = generated_text.split(stop_phrase)[0].strip()

return generated_text
```

After this step, the response breaks after first paragraph. However, this is a post inference step. So, it doesn't reduce the inference time.

Evaluation:

1. Explain what a Small Language Model is.
2. Describe the benefits of SLMs in edge devices.
3. What is the difference between SLM and LLM?
4. Why is temperature important in text generation?
5. Applications of small language models in industry.

Prompt	Perplexity (Original)	Perplexity (With Fix)
1	38.5	34.2
2	40.1	33.8
3	39.7	34.0
4	41.3	33.5
5	38.9	32.9